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Simple view of second language reading: A meta-analytic structural equation modeling approach

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Abstract

Purpose: The present study aimed to systemically summarize the structural relationships among correlated components of second language (L2) reading comprehension to investigate the extent to which the two major components – language comprehension abilities and decoding skills – could account for reading comprehension in L2 contexts in accordance with the model of Simple View of Reading.

Method: We used a meta-analytic structural equation modeling. This study included 81 samples (from 67 studies) including 10,526 participants, and the collected dataset successfully fitted the proposed model of Simple View of L2 Reading.

Results: L2 comprehension abilities and L2 decoding skills accounted for over 60% of the variation in L2 reading comprehension, with the former having a larger contribution. Considering age and L2 proficiency as moderator variables, it was revealed that L2 decoding skills played less important roles for more proficient and older learners, whereas L2 comprehension abilities maintained their importance across different ages and L2 proficiency levels. The moderators related to language differences between learners' first language and L2 did not show significant moderating effects.

Conclusion: The study presented acceptable model fit indices for its models. Future studies can incorporate more pooled correlation coefficients and variables to investigate an effective instruction for L2 reading.

Introduction

The aim of the present study was to systematically summarize the structural relationships among correlated components of second language (L2) reading comprehension using meta-analytic structural equation modeling (MASEM). The model of Simple View of Reading (SVR; Gough & Tunmer, 1986) has been tested and validated by multiple empirical studies with not only monolingual (for a meta-analysis, see, Quinn & Wagner, 2018), but also bilingual students; these studies have investigated if the model can also sufficiently account for the reading comprehension of bilingual students, such as English language learners in the US (e.g., Beattie, 2018; Hoover & Gough, 1990). For example, English as a second language (ESL) learners tend to have lower reading comprehension skills than their peers whose native language is English (e.g., August et al., 2005); hence, understanding the structure of their L2 reading comprehension skills is of importance for educators to assist them in improving their reading comprehension using more effective and efficient means (e.g., Nakamoto et al., 2007; Proctor et al., 2005). For this reason, consistent empirical efforts have been undertaken to understand the structural relationships among correlated components of L2 reading comprehension (e.g., Jeon & Yamashita, 2014; Newell et al., 2020). Accordingly, there is also a strong need to synthesize these findings to identify overall structural relations from an accumulated, and thus, more reliable viewpoint.

The MASEM approach is a relatively new statistical approach in the field of language education, and can be described as a combination of meta-analysis and structural equation modeling. Meta-analysis is gaining popularity in the field (e.g., Jeon & Yamashita, 2014; Lee et al., 2019) as this approach enables researchers to synthesize previous quantitative findings by not only estimating an average effect size across different studies, but also explaining the differences across studies through the moderator analyses. Structural equation modeling is another popular statistical approach, which allows researchers to investigate complex relations between multiple observed and latent variables within a hypothesized model (Kline, 2005). In sum, MASEM is a statistical technique for researchers to answer more intricate research questions by fitting a hypothesized model with multiple observed and latent variables based on a pooled correlation matrix coming from a synthesis of previous study results (Jak & Cheung, 2020; Peng et al., 2021). The present study has been inspired by recent large-scale MASEM-based studies on SVR by Peng et al. (2021), and Quinn and Wagner (2018), which concerned first language (L1) readers. However, the field has still not seen an attempt that systematically synthesizes the results of the studies based on L2 contexts, in which L2 learners are under pressure to develop their L2 reading comprehension abilities at a fast pace to keep up with the

curriculum, which is mostly conducted in the L2. Clearly, there is a strong need to examine the structural relationships among correlated components of L2 reading comprehension in such a context.

To this end, in the present study, we aim to present the model of simple view of second language reading (SV2R) using a MASEM approach to synthesize extant study findings on L2 reading comprehension. In the remainder of the paper, we review the background to SV2R and potential moderators. Then, we describe our research method, followed by the results. Finally, we present an in-depth discussion of the findings and limitations of the study. We expect the present MASEM study to deepen our understanding of L2 reading comprehension and of this relatively new and powerful statistical method. The following research questions guided the present study.

Research Question 1. When using the MASEM approach, do the coefficient correlations from previous literature statistically support the SVR-based model in L2 (i.e., SV2R)?

Research Question 2. To what extent do the identified moderators (i.e., learners' age and L2 proficiency, L1-L2 difference) influence the relationships between L2 reading comprehension and its two sub-components, namely, L2 decoding skills and L2 language comprehension abilities?

Background

The model of SVR

SVR is a theoretical model of reading comprehension, initially proposed by Gough and Tunmer (1986). This model has two central propositions: (1) reading comprehension is influenced by two major components, namely language comprehension abilities and decoding skills, which make unique and shared contributions to reading comprehension; (2) these components are necessary for successful reading comprehension. Language comprehension abilities have been generally considered to be related to oral language proficiency, with listening comprehension often being measured for this construct (Gough & Tunmer, 1986; Hoover & Gough, 1990). Decoding skills are often described as *efficient word recognition*, which is "the ability to rapidly derive a representation from printed input that allows access to the appropriate entry in the mental lexicon, and thus, the retrieval of semantic information at the word level" (Hoover & Gough, 1990, p. 130). Regarding the relationship between these two components, studies have tested multiple perspectives. The original proponents of the SVR (Hoover & Gough, 1990), based on the data from Grades 1 to 4 bilingual readers, argued for a multiplicative model over an additive one, suggesting that the former could predict most of the variances in reading comprehension. On the other hand, based on a study of readers

with English as L1, Chen and Vellutino (1997) proposed a more complicated model (a "weaker" version in their own words) that included "both the linear and multiplicative combinations of decoding and language comprehension" (p. 3) in the equation. The nature of the relationship between these two components has not yet been determined, with some studies having tested a revised formula with an additional component (e.g., speed of processing; Adlof et al., 2006; Joshi & Aaron, 2000) or with another population (e.g., Chinese readers; Yeung et al., 2016). Nevertheless, these studies have all identified the contribution of language comprehension abilities and decoding skills as independent and significant in reading comprehension.

The SVR model has indeed stood the test of time since its inception, with a considerable amount of research attesting to its validity and predictability in L1 contexts (e.g., Catts et al., 2005; Georgiou et al., 2009; Joshi & Aaron, 2000). Research on SVR and its components has been extended to studies with L2 learners or bilinguals, including English-Spanish bilingual children (e.g., Goodwin et al., 2015; Gottardo & Mueller, 2009; Hoover & Gough, 1990) and those who have learned Dutch as an L2 (e.g., Droop & Verhoeven, 2003; Verhoeven & van Leeuwe, 2012), among others. The results of these studies, albeit with some differences in the instruments employed to measure the target components, suggest that language comprehension abilities and decoding skills are important predictors of L2 reading comprehension.

The present study focuses on reading comprehension among learners in L2 (rather than in foreign language learning) environments; hence, it is worth addressing the findings of some of the SVR-based studies conducted in this context (e.g., Bowyer-Crane et al., 2017; Gottardo et al., 2018; Gottardo & Mueller, 2009; Mancilla-Martinez & Lesaux, 2010; Proctor et al., 2005; Uchikoshi, 2013), a majority of which included Spanish-speaking English learners. Overall, the findings of these studies have confirmed that the SVR model can account for a significant amount of variance in L2 reading comprehension as well, revealing its wide applicability for L2 learner-population. For example, the results of Gottardo and Mueller's (2009) study showed that English language proficiency and word reading abilities, which are on par with decoding abilities in the original model, could significantly and strongly predict English reading comprehension of Spanish-speaking English learners at the early stage of English literacy. Similarly, language comprehension abilities and decoding skills are two major components that contribute to reading comprehension, and these relationships have been reported to be sensitive to some moderators, to which we turn next.

Potential moderators

Learner factors

A recurrent finding in the previous studies on SVR is related to the relative importance of language comprehension abilities and decoding skills in reading comprehension as a function of learners' age, with the influence of decoding skills gradually decreasing (e.g., Lonigan et al., 2018; Tilstra et al., 2009; Verhoeven & van Leeuwe, 2012). A meta-analysis of 110 studies on English readers (García & Cain, 2014) confirmed this proposition, revealing that "the relationship between decoding and reading comprehension decreased with increasing chronological age" (p. 92). Thus, we can say that age is a moderator that must be proven by the model.

Along with age, language proficiency has been examined to have a significant effect on the role of decoding skills in reading comprehension; that is, decoding skills have been suggested to have a significantly greater impact for those with a lower level of language proficiency (Catts, 2018; García & Cain, 2014). The decreasing degree of decoding skills' contribution to reading comprehension over time appears to be related to the premise that decoding is a component that becomes faster and more automatic as the reader progresses and develops their language proficiency (Catts, 2018); hence, it plays a less important role at later stages. In the realm of L2 reading research, Grabe (2009) also pointed to the role of language proficiency, such that L2 readers with lower levels of L2 proficiency tend to rely on local strategies, which are more concerned with word-level reading (including decoding).

Overall, literature has implicated the potential roles of age and language proficiency, which are often correlated in several pedagogical contexts, in moderating the contributions of language comprehension abilities and decoding skills to the reading comprehension of monolingual readers; the present study aimed to examine if such implications could be extended to L2 contexts.

L1-L2 difference

A research branch known as Contrastive Analysis (CA; Odlin, 1989; Stockwell et al., 1965) has presented a method of analyzing the structures (e.g., syntax and phonology) of the L1 and L2, and proposed that the degree of structural similarities (or differences) between the two languages may positively or negatively influence L2 learning. This branch has contributed to the field of L2 acquisition by identifying types of L2 errors that a learner-group with a particular L1 would produce, as well as the potential sources for negative transfer.

In light of the broader framework of CA research, as well as studies focusing on the L1-L2 difference in L2 reading (e.g., Bernhardt, 2011), Jeon and Yamashita (2014) included "L1-L2 language distance" (i.e., whether both L1 and L2 are Indo-European languages or only one of them is) and "L1-L2 script distance" (i.e., whether both L1 and L2 use alphabetic writing systems or only one of them does) as variables that may moderate the relationships between L2 reading comprehension and its correlates. Their meta-analysis revealed that L1-L2 language distance did not have a significant moderating effect on the relationship between L2 reading comprehension and L2 linguistic knowledge (i.e., vocabulary and grammar). Similarly, L1-L2 script distance did not show statistically significant moderating effects on the relationship between L2 reading comprehension and decoding skills.

In defining language differences, however, we believe that the classification system based on regions (i.e., language distance) or orthographic systems (i.e., script difference) may have some limitations. For example, Chinese and Japanese are not Indo-European languages, but they are different in several aspects, including their orthographic systems, grammar, and pronunciation. Moreover, being non-alphabetic does not make two orthographic systems similar. As part of an effort to adopt a finer classification to improve the similarity groupings, Beaufils and Tomin (2020) suggested a stochastic approach based on computer and statistical methods to worldwide language classification, which can provide quantifiable values of language differences, such as 1–30: highly related; 30–50: related; 50–70: remotely related; 70–78: very remotely related; and 78–100: not related. The precision of this approach, which was based on a word list, sound change rules, and other statistical parameters to build the stochastic model, matches that of Jäger's (2018) language classification results based on a clustering approach with word lists of worldwide languages, as well as human-based reference classifications.

It should also be noted that Jeon and Yamashita (2014) examined the moderating effects of L1-L2 language distance and L1-L2 script distance on the bivariate relationships only among individual components, owing to the analysis method adopted. As we adopted a MASEM approach and Beaufils and Tomin's (2020) quantified language difference index, the current study examined the extent to which this moderator influences the relationships between the target constructs within an overall framework of an SVR-based model.

The present study: MASEM for SV2R

The present study aimed to synthesize the findings of previous empirical L2 studies on the simple view of reading using MASEM to enable us to present the SVR-based model of reading comprehension in L2 contexts, i.e., the SV2R. It should be clarified here that we have taken the commonly seen empirical approach to fit a model of reading comprehension

accounted for by correlated decoding and language comprehension abilities. That is, we are not fitting a model according to the original SVR theory, which posited a multiplicative relationship, which can be modeled as an interaction in statistical terms. As part of meta-analysis research, MASEM is useful for investigating multivariate relationships between more than two variables by fitting a structural equation modeling (SEM) model using a pooled correlation matrix based on the collected raw correlation coefficients from previous studies (Jak & Cheung, 2020). The following steps were undertaken to build our hypothesized MASEM model of L2 reading comprehension: First, large-scale MASEM studies on SVR in L1 contexts (Peng et al., 2021; Quinn & Wagner, 2018) were consulted to identify an evidence-based typical structure for SVR. According to the MASEM models in these studies, *vocabulary* and *listening comprehension* were identified as sub-components of language comprehension abilities, while *fluency* and *accuracy* were identified as sub-components of decoding skills. Second, previous SVR-based studies in L2 contexts (e.g., Bowyer-Crane et al., 2017; Gottardo et al., 2018; Gottardo & Mueller, 2009; Mancilla-Martinez & Lesaux, 2010; Proctor et al., 2005; Uchikoshi, 2013) as well as recent meta-analyses on L2 reading comprehension (e.g., Jeon & Yamashita, 2014; Zhang & Zhang, 2022) were perused to identify and narrow down the list of correlates of L2 reading comprehension within the SVR framework. Consequently, for a possible structure of SV2R, the following variables were identified as sub-components of language comprehension abilities: vocabulary knowledge (e.g., Gottardo & Mueller, 2009; Uchikoshi, 2013), grammatical (or syntactic) knowledge (e.g., Bowyer-Crane et al., 2017; Gottardo et al., 2018), and listening comprehension (e.g., Bowyer-Crane et al., 2017; Uchikoshi, 2013). Fluency and accuracy were identified as the sub-components of decoding skills (e.g., Mancilla-Martinez & Lesaux, 2010; Proctor et al., 2005). This combination of fluency and accuracy, also known as decoding efficiency, has been frequently used to measure learners' decoding skills (e.g., Beattie, 2018; Lervåg & Aukrust, 2010; Lesaux et al., 2010). Based on the aforementioned preliminary identification of the potential sub-components and original SVR model, we presented our hypothesized SV2R model, as illustrated in Figure 3.

Methods

Literature search

As part of a regular meta-analytic approach, we conducted keyword database search, journal search, and backward/forward reference search of review articles and identified articles. For the database search, we searched articles published till December 2020, and used the following keywords: *second language reading* AND *listening*, *language comprehension*,

vocabulary, *grammar*, *decoding*, *accuracy*, and *fluency*, using Web of Science, SCOPUS, and ProQuest (including Dissertation & Theses Global). For the journal search, we conducted keyword searches in the following journals: *Applied Linguistics*, *Applied Psycholinguistics*, *Child Development*, *Educational Research Review*, *Foreign Language Annals*, *Journal of Educational Psychology*, *Journal of Research in Reading*, *Reading and Writing*, *Review of Educational Research*, and *Scientific Studies of Reading*. For the last step, we reviewed reference lists of review articles and identified articles to determine relevant studies cited by these articles (i.e., backward search). We also conducted a forward search using Google Scholar's "Cited by" function to locate any relevant studies that cited back to the review articles and previously identified articles.

After removing duplicates from the pool of studies identified via multiple resources, we began the screening process. The first and third authors of the present study sat beside each other and identified 82 articles out of 164 as irrelevant to the topic after the title/abstract screening. Furthermore, the two authors located PDF files of the 82 studies and reviewed them together to check whether they met the inclusion criteria, including the following: (1) studies should have been conducted in an L2 context; (2) studies should have reported original raw correlation coefficients (*r*) between passage-level L2 reading comprehension and its correlates; (3) studies should have reported separate data in case both monolingual and L2 readers (learners) were included as participants; (4) studies should have reported the sample size; and (5) studies should have been written in English. Figure 1 presents the flow chart of the literature search.

Coding procedure

Coding scheme

As can be seen from Figure 1, we included 67 studies after excluding 15 studies because two studies were not conducted in L2 contexts, six studies were not reading-related studies, one study did not separate the data between monolingual and L2 learners, and six studies did not report original raw correlation coefficients. When reviewing the collected 67 studies (see *Online supplemental file* for the list of the studies included in our analysis), we found that some of them had more than one independent sample; therefore, we identified a total of 81 samples with 10,526 participants. These samples were then coded to build a dataset that consisted of correlation matrices, sample sizes, and moderators, such as learner factors (i.e., age, L2 proficiency), and L1-L2 difference.

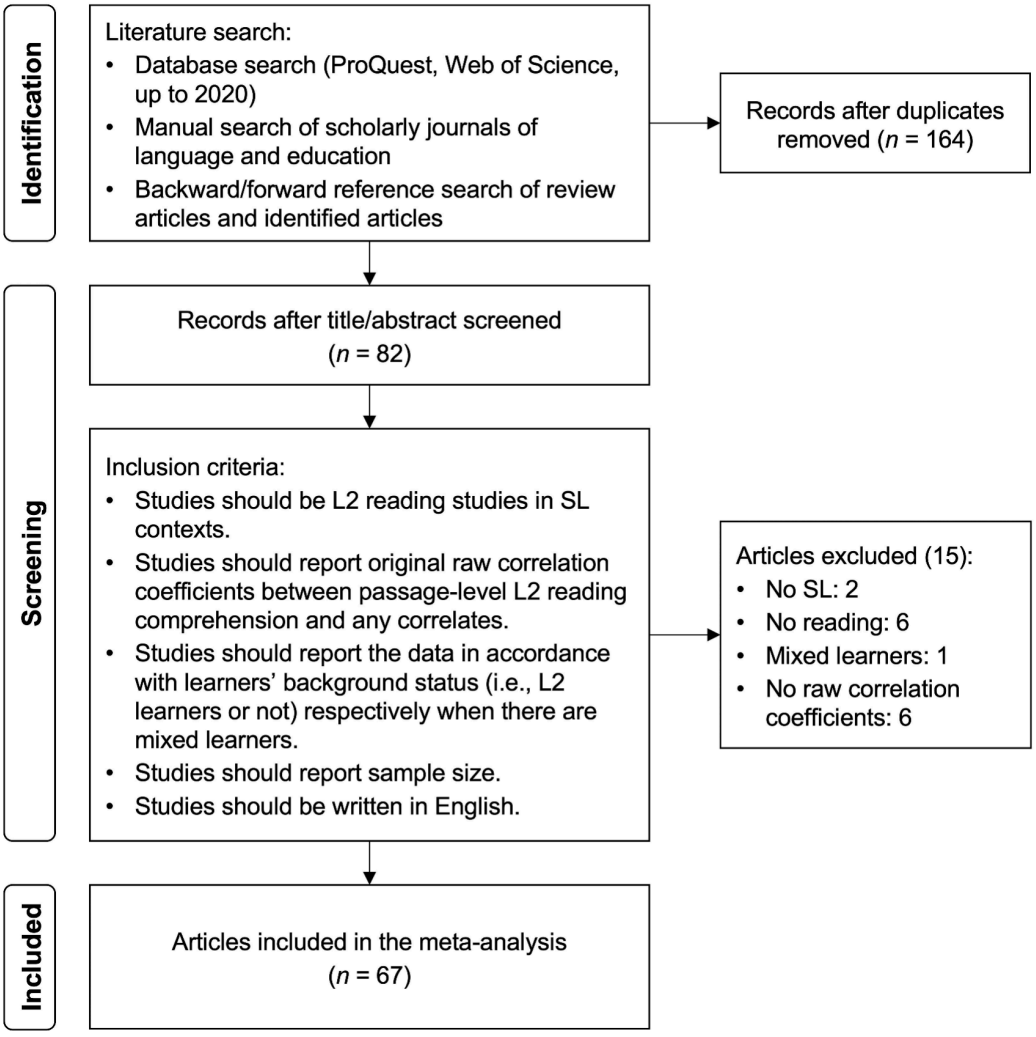


Figure 1. PRISMA flow chart of the literature search (Page et al., 2021).

Correlation matrices

For the correlation matrices, we first transcribed all the reported raw correlation coefficients among various measurements for each identified sample. We then reviewed the identified measurements to categorize them into the seven target variables: (1) L2 reading comprehension, (2) L2 vocabulary knowledge, (3) L2 grammar knowledge, (4) L2 listening comprehension, (5) L2 decoding efficiency, (6) L2 decoding accuracy, and (7) L2 decoding fluency. The definitions and examples of the measures for the seven target variables are presented in Table 1. When there were multiple correlations for a pair of variables, such as in cases where multiple assessments had been conducted (e.g., for both vocabulary size and vocabulary depth) or assessments had been conducted multiple times (e.g., pretest and posttest), we averaged those correlations to control for data dependency (for a similar approach, see, Peng et al., 2021; Zhang & Zhang, 2022).

Moderator variables

For the moderator variables, we first transcribed all information that was relevant to learner factors (i.e., age, L2 proficiency), and L1-L2 difference. We then reviewed the information to categorize it into the values for each moderator. The definitions used in Jeon and Yamashita's (2014) were referred to, such that: (1) age was coded as child (those who are in or below Grade) or adolescent/adult, (2) L2 proficiency was coded as basic (those who are beginner level students or in the 1st year of L2 learning) or beyond basic. L1-L2 difference was coded as not related (78–100) or related (1–77), according to the results of Beaufils and Tomin's (2020) genetic proximity calculator between two languages.

As expected, each sample had various types of measurements, requiring decisions regarding how to categorize these measurements into our correlation matrix template with the seven target variables. The definitions of the variables of interest (as listed in Table 1) were discussed until a consensus was reached among the three authors of the present study. In contrast, coding the moderator variables was relatively uncontentious except for unclear information in the original articles. The three authors independently coded 33% of the studies each, and the first author reviewed all the transcribed correlation matrices and moderator variables to ensure interrater reliability. A few disagreements between the authors were resolved by discussion and the authors reached a full consensus across the total correlation matrices. Table 2 presents the final list of variables and the number of correlations among them, Table 3 presents the descriptive statistics for the three moderator variables, and Table 4 presents detailed information about the moderator variable related to L1-L2 difference.

Table 1. Definitions and examples of the target variables.

Constructs	Variables	Definitions	Examples
L2 reading comprehension		Tasks that measure the ability to comprehend the passage or text	Passage- or text-level reading comprehension test
L2 comprehension abilities	L2 vocabulary knowledge	Tasks that measure students' multidimensional lexical knowledge	Vocabulary size test, depth test
	L2 grammar knowledge	Tasks that measure the ability to analyze and integrate syntactic information	Syntactic awareness test, cloze test
	L2 listening comprehension	Tasks that measure the ability to comprehend spoken language	Listening or oral language comprehension test
L2 decoding skills[a]	L2 decoding efficiency	Tasks that measure the ability to read words accurately and fluently	Word reading efficiency (e.g., Test of Word Reading Efficiency)
	L2 decoding accuracy	Tasks that measure the ability to read words accurately	Word reading accuracy, word-letter identification
	L2 decoding fluency	Tasks that measure the ability to read words fluently	Word reading fluency

[a]Efficiency and fluency measures for the decoding component did not include any measures that involve text reading (e.g., oral reading fluency [ORF] measures) or comprehension (e.g., Test of Silent Reading Efficiency and Comprehension [TOSREC]).

Table 2. Final list of variables and the number of correlations among them.

Variables	1	2	3	4	5	6	7
1: L2 reading comprehension	81 (10,526)						
2: L2 vocabulary knowledge	58 (8,188)	81 (10,526)					
3: L2 grammar knowledge	23 (3,069)	16 (2,709)	81 (10,526)				
4: L2 listening comprehension	25 (4,015)	13 (2,056)	3 (890)	81 (10,526)			
5: L2 decoding efficiency	21 (2,731)	15 (1,783)	6 (685)	9 (976)	81 (10,526)		
6: L2 decoding accuracy	25 (3,784)	19 (2,225)	5 (479)	8 (1,434)	9 (1,031)	81 (10,526)	
7: L2 decoding fluency	23 (2,700)	17 (2,227)	6 (1,075)	8 (1,372)	7 (794)	4 (367)	81 (10,526)

Values in parentheses indicate total sample sizes for correlation coefficients collected from primary studies.

Table 3. Descriptive statistics for moderator variables.

Moderators	Categories	Values
Age	Child	58 out of 81 (72%)
	Adolescent/Adult	23 out of 81 (28%)
L2 proficiency	Basic	33 out of 81 (41%)
	Beyond basic	48 out of 81 (59%)
L1-L2 language difference	Related	35 out of 55 (64%)
	Not related	20 out of 55 (36%)

26 samples did not have relevant information about L1-L2 difference.

Table 4. Detailed information about L1-L2 difference (from 55 samples).

Number of samples		L1	L2	Genetic proximity between L1 and L2 (78–100: no relation; 70–78: very remotely related; 50–70: remotely related; 30–50: related; 1–30: highly related)*		Country
39	2	Cantonese	English	90.1	Not related	US
	4	Chinese		82.4	Not related	
	4	Japanese		88.3	Not related	
	1	Korean		90.0	Not related	
	28	Spanish		57.0	Related	

Number of samples		L1	L2	Genetic proximity between L1 and L2 (78–100: no relation; 70–78: very remotely related; 50–70: remotely related; 30–50: related; 1–30: highly related)*		Country
6	1	Portuguese	English	59.8	Related	Canada
	1	Farsi		78.0	Not related	
	1	Chinese		82.4	Not related	
	1	Spanish	French	37.1	Related	
	2	English		48.7	Related	
2	1	Chinese	English	82.4	Not related	UK
	1	Japanese		88.3	Not related	
2		Russian	Hebrew	88.9	Not related	Israel
2	1	French	Dutch	62.4	Related	Belgium
	1	Dutch	French	62.4	Related	
1		Turkish	Dutch	96.5	Not related	Netherlands
2	1	Chinese	Japanese	81.5	Not related	Japan
	1	Korean		88.0	Not related	
1		Urdu	Dutch	71.7	Related	Norway

*Genetic proximity between languages was measured by an automatic language classification approach based on computer and statistical methods presented by Beaufils and Tomin (2020).

Missing data

The samples had missing data for both correlation coefficients and moderator variables. First, for each of the correlations for a pair of variables (as shown in Table 2), the extent of missing data ranged from 3.7% (3 missing out of 81) to 71.6% (58 missing out of 81), and our MASEM approach used the full information maximum likelihood (FIML) estimation method (Jak & Cheung, 2020) to handle the missing correlation estimates (for a similar approach, see, Peng et al., 2021). Assuming that the data are missing at random or completely at random, indicating that the researcher did not include the correlation without a specific reason (e.g., they probably had no interest in that correlation), the FIML method works by estimating model likelihood function from a summation across estimated likelihood functions for each individual case based on the available sample data (Cham et al., 2017). In this way, instead of imputing the values of missing data or deleting variables with missing data, one can estimate a likelihood function with the available data. Second, with respect to the moderator analysis, we used a listwise deletion method to deal with the missing data – we simply dropped the variables with missing data (for a similar approach, see, Peng et al., 2021). For the learner factors (i.e., age, L2 proficiency), we did not drop any case. However, for the moderators related to L1-L2 difference, 26 samples were dropped from the dataset because of missing

data, and this affected the number of correlation coefficients among the target variables. Therefore, we had to exclude the L2 grammar variable (n = 0) from the model (i.e., Model 2), as illustrated in Figure 5.

Data analysis plan

We implemented a recent MASEM approach called the one-stage MASEM, suggested by Jak and Cheung (2020; its statistic package is included in the "metaSEM" *R* package Version 1.2.5.1 provided by Cheung, 2015), using *R* Version 4.0.3. This approach is gaining attention from the research community (e.g., Peng et al., 2021) as it allows researchers to include moderators in the model such that they can directly evaluate the extent to which moderators influence the relationships between constructs in the MASEM model (for more details, see, Jak & Cheung, 2020).

First, using the MASEM approach, we built a pooled correlation matrix based on the 81 correlation matrices from each sample. Second, using the computed pooled correlation matrix, we estimated our hypothesized SEM model for SV2R (Model 1; 81 samples; N = 10,526), as shown in Figure 3. To confirm a successful fit between our dataset and the hypothesized model, we carried out the SEM goodness-of-fit test (i.e., χ^2 test; either $p > .05$ or chi-square to df ratio [χ^2/df] ≤ 5 for acceptable fit) and checked four widely used SEM model fit indices (e.g., Kline, 2005; Lee et al., 2020): (1) root mean square error of approximation (RMSEA; < .08 for acceptable fit), (2) root mean square residual (SRMR; < .08 for acceptable fit), (3) comparative fit index (CFI; > .90 for acceptable fit), and (4) Tucker–Lewis index (TLI; > .95 for acceptable fit).

Second, after estimating the hypothesized models, we conducted moderator analyses. These analyses were conducted to gain an understanding of how moderator variables influence the relationships between L2 reading comprehension and the two components: L2 comprehension abilities and L2 decoding skills. We conducted the first moderator analysis by including each of the learner related moderators, i.e., age (child vs. adolescent/adult) and L2 proficiency (basic vs. beyond basic), to Model 1. Prior to evaluating the impact of the remaining moderator variable, i.e., L1-L2 language difference, we dropped the L2 grammar variable owing to the reason stated above (i.e., Model 2; 55 samples; N = 5,511), and then used this model for the second moderator analysis.

Third, to examine the statistical significance of the moderating effects, we compared the models without a moderator to the ones with the moderator with likelihood ratio statistic. The effects of the moderators were deemed significant when $p < .05$ in log-likelihood.

Publication bias

We conducted exploratory analyses for the publication bias testing, as, to the best of our knowledge, there are no known methods to investigate publication bias within the MASEM framework. Therefore, we adopted the methods suggested by Peng et al. (2021).

First, we analyzed the reporting differences (i.e., the number of reported correlation data points for the correlation matrix template of the present study) across publication types (i.e., gray literature vs. published articles). Out of the 81 samples, four (5%) came from dissertations while the remaining 77 (95%) came from published articles, reporting an average of 5.50 (*SD* = 3.32) and 3.87 (*SD* = 3.34) correlation data points, respectively. The results of the *t*-test revealed no statistically significant difference between the average numbers of reported correlation data points ($t = 0.95$, $SE = 1.71$, $p = .34$).

Second, we conducted a moderator analysis by using publication type as the moderator in Model 1. The likelihood ratio test ($\chi^2(2) = 1.03$, $p = .60$) suggested that publication type did not have any significant influence on the paths of interest in the present study.

Third, we conducted Egger's test to predict the 21 computed pooled correlation coefficients using their standard errors to check whether any of the calculations for correlation coefficients between the variables of interest were beyond their normal range. The results indicated no significant sign of publication bias ($z = -1.02$, $p = .31$), and all coefficients were located within pseudo 95% confidence limits in the funnel plot (Figure 2).

In conclusion, these exploratory publication bias testing approaches indicated that there was no significant publication bias among the collected data.

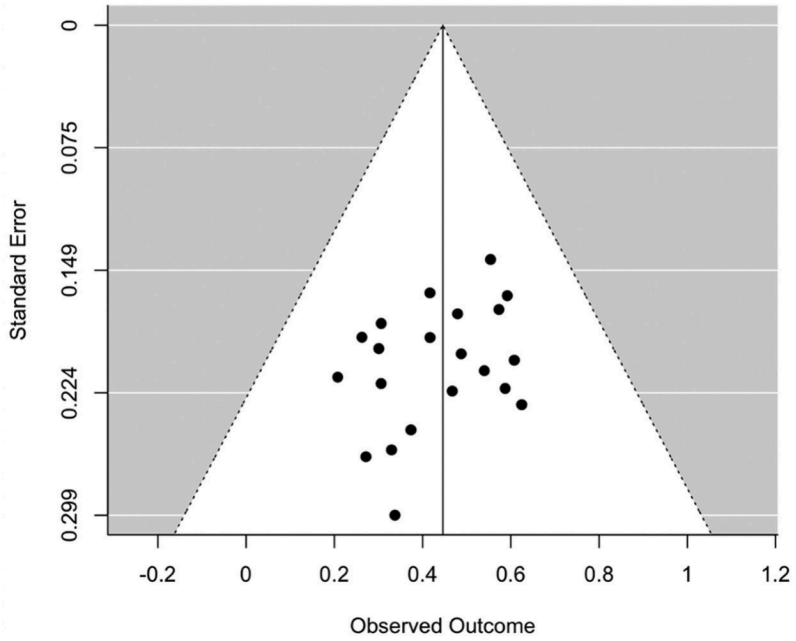


Figure 2. Funnel plot for the computed 21 pooled correlation coefficients and their standard errors.

Results

Model 1: overall analysis

First, we estimated a pooled correlation matrix among the seven variables included in the current meta-analysis (81 samples with 10,526 participants). As shown in Table 5 (below diagonal), a total of 21 pooled correlation coefficients among the variables ranged from .21 to .62, with all correlations being statistically significant ($p < .05$).

Based on the pooled correlation matrix, we built an SEM model with L2 reading comprehension as the dependent variable and two latent variables (i.e., constructs) as the independent variables. As shown in Figure 3, the two latent variables in Model 1 were supported by significant and strong factor loading coefficients (βs > .60). Moreover, we found that the dataset we built supported our hypothesized model (Model 1) with an overall acceptable model fit: $\chi^2(12) = 23.73$, $p = .02$, $\chi^2/df = 1.98$; CFI = .98; TLI = .97; RMSEA = .01; and SRMR = .07.

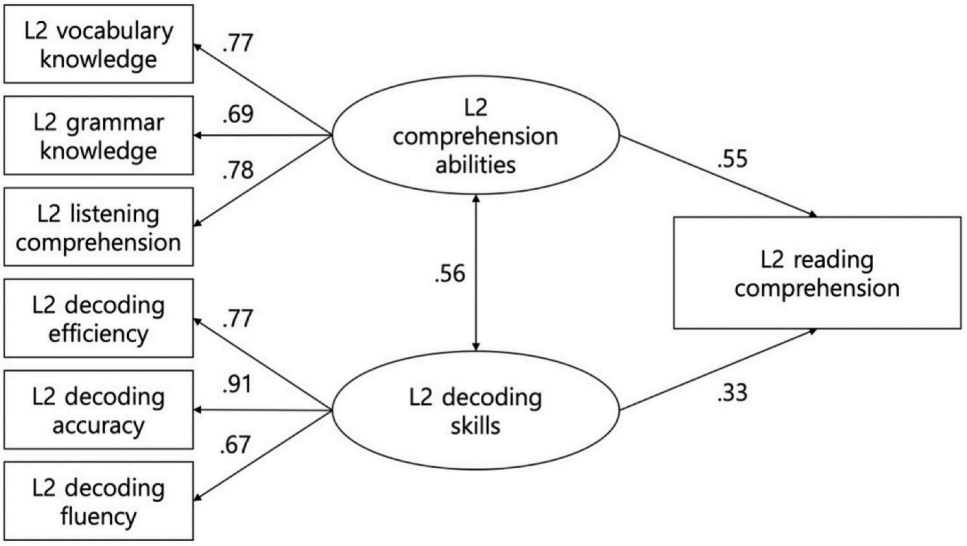


Figure 3. The hypothesized MASEM model for SV2R with estimates (Model 1). The solid lines in the figure represent the statistically significant path coefficients.

Table 5. Pooled correlation matrix for the current MASEM approach (Models 1 and 2).

Variables	**1**	**2**	**3**	**4**	**5**	**6**	**7**
1: L2 reading comprehension	-	.51 (35)	NA	.60 (18)	.48 (18)	.57 (23)	.41 (14)
2: L2 vocabulary knowledge	.55 (58)	-	NA	.67 (8)	.28 (12)	.39 (17)	.24 (11)
3: L2 grammar knowledge	.49 (23)	.54 (16)	-	NA	NA	NA	NA
4: L2 listening comprehension	.59 (25)	.61 (13)	.37 (3)	-	.33 (8)	.47 (7)	.11 (5)
5: L2 decoding efficiency	.48 (21)	.31 (15)	.33 (6)	.31 (9)	-	.63 (9)	.53 (6)
6: L2 decoding accuracy	.57 (25)	.42 (19)	.27 (5)	.47 (8)	.62 (9)	-	.33 (4)
7: L2 decoding fluency	.42 (23)	.26 (17)	.30 (6)	.21 (8)	.59 (7)	.34 (4)	-

All correlations for Model 1 (below diagonal) and Model 2 (above diagonal) are statistically significant, $p < .05$. Values in parentheses indicate the number of raw correlation coefficients from the included studies.

As can be seen in Figure 3, the results for Model 1 indicate a significant correlation between L2 comprehension abilities and L2 decoding skills ($\beta = .56$), both of which contribute significantly to L2 reading comprehension (β for L2 comprehension abilities → L2 reading comprehension = .55, β for L2 decoding skills → L2 reading comprehension = .33). Overall, we found that Model 1 explained 61% variance in L2 reading comprehension. To compare the magnitudes of these two path coefficients, we conducted a likelihood ratio (LR) test. The results indicated a significant difference between the path coefficients: L2 comprehension abilities contributed more to L2 reading comprehension than L2 decoding skills ($\chi^2(1) = 4.44$, $p = .04$).

Model 1: moderation analysis for learner factors

After checking the model fit of our hypothesized model, we conducted the first moderator analysis to investigate how learners' age and L2 proficiency influence the path coefficients between the constructs. For age and L2 proficiency, we included the moderator variables into Model 1, and estimated Model 1a and Model 1b as shown in Figures 4(a) and 4(b), respectively. Above all, we found that both Models 1a and 1b were significantly different from Model 1 according to the likelihood ratio test results ($\chi^2(2) = 6.61$, $p = .04$; $\chi^2(2) = 19.20$, $p = .00$; respectively). Therefore, we concluded that age and L2 proficiency had significant moderating effects on the relationship of L2 reading comprehension with L2 comprehension abilities and L2 decoding skills.

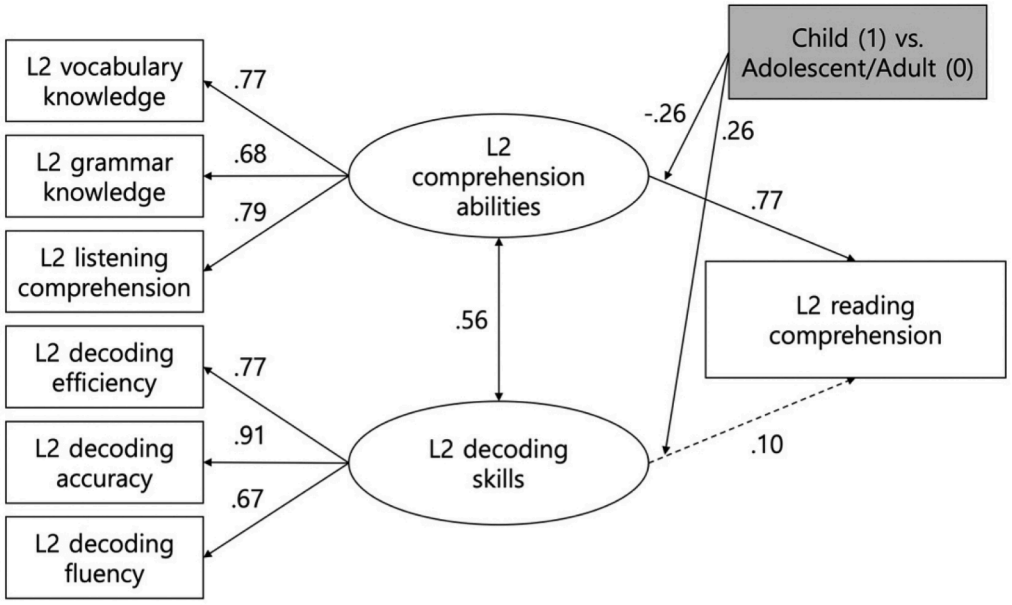
In particular, the two path coefficients between each of the two latent variables and the L2 reading comprehension variable shown in Figures 4(a) and 4(b) are for learners with categorical values of zero for the moderator variables: the adolescent/adult learners (Model 1a) and learners with beyond basic L2 proficiency level (Model 1b). For these learners, the results indicate a more important role of L2 comprehension abilities (i.e., $\beta_{MODEL\ 1} = .55 \rightarrow \beta_{MODEL\ 1A} = .77$ [$\beta = -0.26$, $SE = 0.11$, $p = .01$]; $\beta_{MODEL\ 1B} = .69$ [$\beta = -0.31$, $SE = 0.07$, $p = .00$]) compared to L2 decoding skills (i.e., $\beta_{MODEL\ 1} = .33 \rightarrow \beta_{MODEL\ 1A} = .10$ [$\beta = 0.26$, $SE = 0.14$, $p = .07$]; $\beta_{MODEL\ 1B} = .19$ [$\beta = 0.28$, $SE = 0.07$, $p = .00$]) in determining L2 reading comprehension.

For the child learners (Model 1a) and learners with basic L2 proficiency level (Model 1b), adding the moderating effects to the path coefficients between the latent variables and the L2 reading comprehension variable revealed the less important role of L2 comprehension abilities (i.e., $\beta_{MODEL\ 1} = .55 \rightarrow \beta_{MODEL\ 1A} = .50$ [$\beta = -0.26$, $SE = 0.11$, $p = .01$]; $\beta_{MODEL\ 1B} = .38$ [$\beta = -0.31$, $SE = 0.07$, $p = .00$) compared to L2 decoding skills (i.e., $\beta_{MODEL\ 1} = .33 \rightarrow \beta_{MODEL\ 1A} = .36$ [$\beta = 0.26$, $SE = 0.14$, $p = .07$]; $\beta_{MODEL\ 1B} = .47$ [$\beta = 0.28$, $SE = 0.07$, $p = .00$]) in determining L2 reading comprehension.

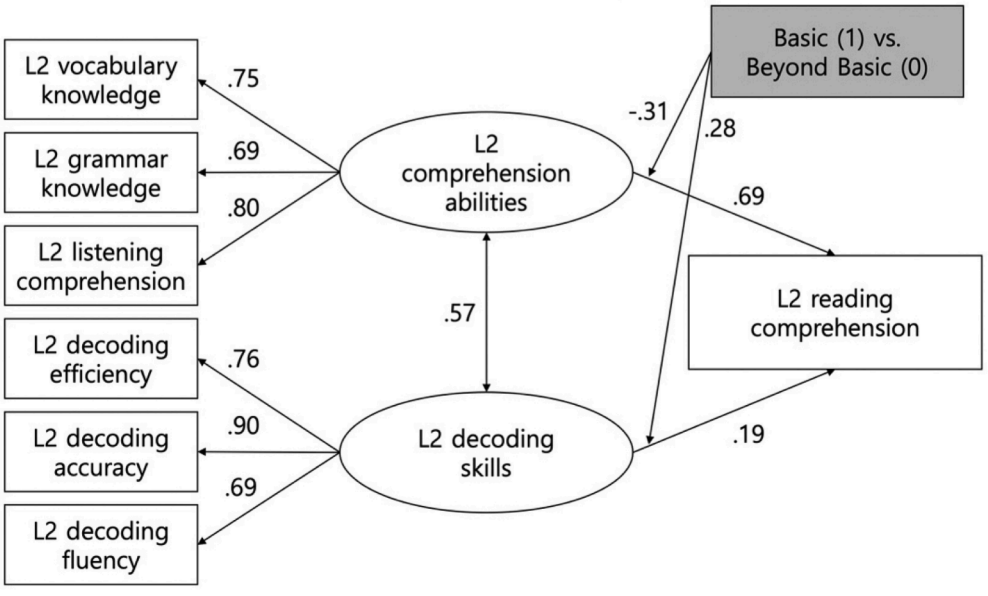
Model 2: overall analysis

In order to estimate moderation effects of L1-L2 difference, we built a revised version of the MASEM model (i.e., Model 2; see Figure 5) after dropping 26 samples, resulting in 55 samples with 5,511 participants. This led to the exclusion of the L2 grammar variable, owing to a lack of available correlation coefficients between this and the other variables. First, we estimated a pooled correlation matrix among six variables. As shown in Table 5 (above diagonal), a total of 15 pooled correlation coefficients among the six variables ranged from .11 to .67, with all correlations being statistically significant ($p < .05$).

Based on the pooled correlation matrix, we then built another MASEM model with L2 reading comprehension as the dependent variable and two latent variables as the independent variables. As shown in Figure 5, the two latent variables in Model 2 (i.e., L2 comprehension abilities and L2 decoding skills) were supported by significant and strong factor loading coefficients (βs > .60). With respect to its goodness-of-fit, we found an overall acceptable fit owing to the following model fit indices: $\chi^2(7) = 19.15$, $p = .01$, $\chi^2/df = 2.74$; CFI = .97; TLI = .94; RMSEA = .02; and SRMR = .08.



(a).



(b).

Figure 4. (a). Model 1a, (b). Model 1b. The solid lines in the figure represent the statistically significant path coefficients.

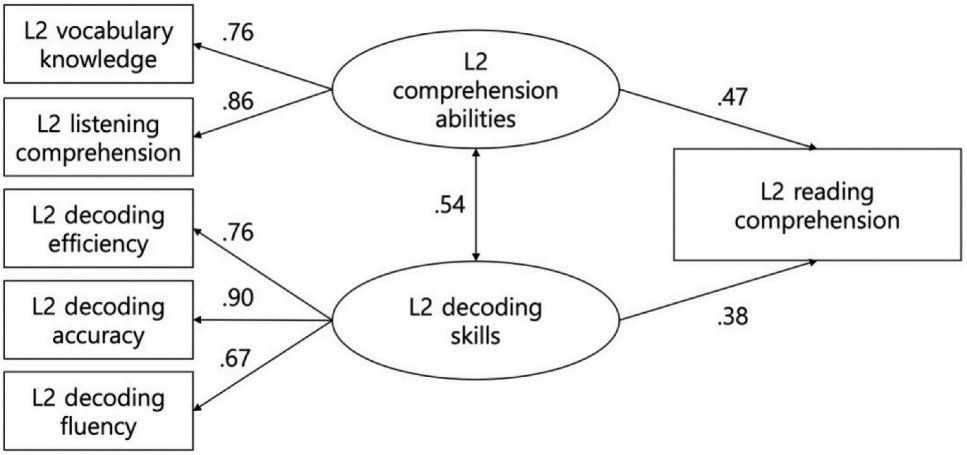


Figure 5. The revised hypothesized MASEM model for SV2R with estimates (Model 2). The solid lines in the figure represent the statistically significant path coefficients.

The results of Model 2 indicated a significant correlation between L2 comprehension abilities and L2 decoding skills (β = .54), with both of these constructs contributing significantly to L2 reading comprehension (β for L2 comprehension abilities → L2 reading comprehension = .47, β for L2 decoding skills → L2 reading comprehension = .38). Overall, we found that Model 2 explained 55.9% variance in L2 reading comprehension. We conducted an LR test to compare the magnitudes of these two path coefficients. The results indicated no significant difference between the path coefficients ($\chi^2(1) = 0.65$, p = .42).

Model 2: moderation analysis for L1-L2 difference

After estimating the revised hypothesized model, we conducted the second moderator analysis to investigate how L1-L2 language difference influences the path coefficients between the constructs. To this end, we included the moderator variable in Model 2 and estimated Model 2a. We found that Model 2a was not significantly different from Model 2 according to the likelihood ratio test results ($\chi^2(2) = 4.75$, p = .09). Therefore, we concluded that the moderator related to L1-L2 language difference did not have significant moderating effects on the relationship of L2 reading comprehension with L2 comprehension abilities and L2 decoding skills.

Discussion

In the present study, we adopted a MASEM approach to present the SV2R model based on the identified studies, comprising 81 samples (N = 10,526). In light of the commonly practiced SVR model, we constructed our model with two correlated major components – L2 comprehension abilities and L2 decoding skills, which, in turn, comprised sub-components

selected on theoretical grounds. In this section, we discuss our findings in reference to the previous studies based on SVR and L2 reading and address the limitations of the study and future directions.

The applicability of SVR in L2 contexts

The first major finding of the present study was that L2 comprehension abilities and L2 decoding skills accounted for over 50% variance in L2 reading comprehension (i.e., 61% in Model 1 including L2 grammar knowledge and 55.9% in Model 2 excluding L2 grammar knowledge). These values allow us to suggest that language comprehension and decoding skills, proposed as necessary components for reading comprehension in the SVR model (Gough & Tunmer, 1986; Hoover & Gough, 1990), play significant roles in L2 reading comprehension. These numbers are also similar to those found in the previous MASEM-based studies on L1 reading comprehension, which ranged from 50% to 60% (Peng et al., 2021; Quinn & Wagner, 2018). In our model, L2 comprehension abilities (β = .55 in Model 1, β = .47 in Model 2) were found to contribute more than L2 decoding skills (β = .33 in Model 1, β = .38 in Model 2), although not significantly in Model 2. This result accorded with the general pattern (i.e., larger or similar degree of contribution of L2 comprehension abilities) observed in the previous MASEM-based studies, such as those by Quinn and Wagner (2018) on L1-English readers and Peng et al. (2021) on L1-Chinese readers, as well as some individual studies with L2 learners (e.g., Bowyer-Crane et al., 2017; Proctor et al., 2005).

In view of this first major finding, the theoretical contribution of the present study is that it confirms and extends the overall findings of the previous MASEM studies based on L1 reading, including the significant roles that language comprehension and decoding skills play in L2 reading comprehension. However, the proportion of unaccounted variance revealed in the present study also indicates a need to examine the role of additional variables in reading comprehension, a trend that has already started to emerge in both L1 and L2 reading studies (see *Limitations and future directions* for further discussion).

The relative importance of L2 decoding skills according to learner factors

The second major finding of the current study was that while the component of L2 comprehension abilities maintained its importance across moderators, the component of L2 decoding skills had relatively different weights for each moderator value. In Model 1 (including L2 grammar knowledge), our results pointed to the significant moderating effects of age and L2 proficiency in terms of the relationships between L2 reading comprehension and the two major components. Regarding proficiency, it was revealed that L2 decoding skills accounted for lesser variance in L2 reading comprehension among those with higher levels of

L2 proficiency. This pattern is briefly summarized in García and Cain's (2014) as follows: "[A]s word reading becomes more fluent and efficient, the variance in reading comprehension by word reading skills will decrease and language comprehension processes will play a more influential role" (p. 96). Thus, the development of L2 decoding skills would make word-level reading automatic (Catts, 2018), resulting in language comprehension abilities becoming more important in later stages of L2 reading comprehension. Indeed, the important role of L2 comprehension abilities in L2 reading comprehension has already been suggested in Jeon and Yamashita's (2014) meta-analysis on L2 reading comprehension and its correlates, which revealed high correlation coefficients between L2 reading comprehension and the sub-components of L2 comprehension abilities (i.e., L2 vocabulary, L2 grammar, and L2 listening).

This pattern continued to remain valid when including age as a moderator variable: the contribution of L2 decoding skills to L2 reading comprehension was smaller for adolescent and adult L2 learners than for young L2 learners (i.e., children). This finding accords with Jeon and Yamashita's (2014) meta-analysis, in which the adult/adolescent L2 groups showed lower correlations between L2 reading comprehension and decoding (r = .46) than the child groups (r = .61). This finding also corresponds with the results of García and Cain's (2014) meta-analysis with L1-English readers and Peng et al.'s (2021) MASEM-based study with L1-Chinese readers, as all three study findings indicate that learners' decoding skills become automatized at some point, following which the contribution of their semantic processing (i.e., language comprehension abilities) to reading comprehension relatively increases.

With these findings in mind, the importance of the development of decoding skills at earlier stages of reading in L1 reading instruction (National Reading Panel, 2000) could be considered to apply in L2 reading as well; hence, learners with limited L2 proficiency level or young L2 learners may benefit from instructions related to word-level reading (i.e., decoding skills; Carrell & Grabe, 2002). However, even with such a focus on decoding skills at early stages of L2 literacy instruction, L2 learners may need to develop their competence in the sub-components of L2 comprehension abilities, which would play a meaningful role collectively at earlier and later stages of L2 reading. For example, it may be important for learners to expand their vocabulary knowledge (e.g., Alderson, 2000; Grabe, 2009; Nation, 2013), which may be related to a lexical coverage rate of reading texts (e.g., Schmitt et al., 2011). Similarly, the advance in L2 grammar knowledge, noted as a key determinant for L2 reading (e.g., Bernhardt, 2011; Grabe, 2009), may be beneficial for L2 learners, as the improvement of such knowledge would facilitate the syntactic processing of the written input in reading texts, which may be fundamental to achieve L2 reading comprehension (Jeon & Yamashita, 2014).

The moderating effects of L1-L2 difference in L2 reading comprehension

The third major finding was related to the effects of L1-L2 language difference on the relationships between L2 reading comprehension and its two major components (i.e., L2 comprehension abilities and L2 decoding skills). L1-L2 language difference did not significantly influence the path coefficients between the constructs. Thus, these results further emphasize the importance of developing one's competence in the components of L2 word recognition (e.g., Carrell & Grabe, 2002), L2 vocabulary (e.g., Droop & Verhoeven, 2003), and L2 grammar (e.g., Bernhardt, 2011), regardless of the differences between learners' L1 and the target language.

The result related to the statistically insignificant role of L1-L2 language difference accords with that of Jeon and Yamashita's (2014) meta-analysis, which revealed no moderating effects of variables related to L1-L2 difference (i.e., L1-L2 language distance and L1-L2 script distance). Despite different operationalization and coding schemes of L1-L2 language difference compared to the present study, these results together indicate that L1-L2 difference may exert effects only on sub-components, such as L2 language comprehension abilities and L2 decoding skills, but these effects do not extend beyond this level. For example, a moderating effect of language similarity and orthographic similarity between L1 and L2, if there is any, may be relevant for L2 language comprehension abilities and L2 decoding skills, respectively, which would be accounted for in the (latent) construct of these elements. Then, the already affected L2 language comprehension abilities and L2 decoding skills would play a major role in forming L2 reading comprehension without being additionally influenced by L1-L2 difference. Thus, this finding lends more weight to the validity of SV2R, in which the variance of L2 reading comprehension is largely accounted for by L2 comprehension abilities and L2 decoding skills (measured in the same language).

Limitations and future directions

The present study is not without limitations. First, although we were able to provide acceptable model fit indices for our MASEM models, having a larger number of pooled correlation coefficients would be more desirable to ensure better model fits. For example, our study included only those studies that were conducted in L2 contexts, wherein the target language is the primary medium of daily communication. The inclusion of studies conducted in foreign language contexts, wherein the target language is rarely spoken outside the classroom environment, will not only broaden our understanding of the applicability of SVR frameworks in L2 reading, but also provide larger samples for a more stable and precise MASEM model. Second, in searching previous studies for the meta-analysis, we included ProQuest's Dissertation and Theses Global to locate gray literature; nevertheless, we could not

reach out to find unpublished studies in the field and those published outside available journals and databases. Not only for a more comprehensive and balanced perspective but also for more precise statistical estimates for data analyses, future meta-analysis research is expected to locate more gray literature than the present study. Third, regarding coding, as mentioned in Method section, two of the authors read all the selected studies carefully and developed the coding scheme and dataset together, as a result of which interrater reliability was not calculated. Although we believe that our close collaboration has resulted in a rigorous coding scheme and dataset, it can prove to be another limitation of the present study, especially when it is more conventional for researchers to undergo a series of training and inter-rater agreement processes that may lead to reduced error rates in coding and may provide a well-developed coding manual so that others can replicate the research. Finally, if we could have secured more samples and greater statistical power, we may have been able to include other relevant variables, which would have helped in expanding the SV2R model by, for example, including higher order cognitive variables (e.g., inference, reasoning; Kim, 2017, 2020) for L2 comprehension abilities and meta-linguistic skills (e.g., phonological and orthographical awareness; Kim, 2017, 2020; Peng et al., 2021) for L2 decoding skills. Overall, the model including these variables is expected to allow for examination of a more comprehensive framework of reading comprehension (Kim, 2020) and eventually contribute to effective instruction for L2 reading.

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